CSE558 · DATA SCIENCE — LECTURE 2

The Vocabulary of Data

Every dataset is objects and attributes. Every attribute has a type. Every type has rules.

TODAY'S ARC

Recape → Attribute types → Encodings → Central tendency

Where we're headed

By the end of today, you'll be able to look at any column of data, name its type, and know exactly which statistics are even valid to compute on it.

RECAP

Where we left off

1

The data science process

Understand → Collect → Prepare → Train & Validate → Insight. Snow ran it by hand in 1854 — a doctor, a map, and a pump handle.

2

Your project

Groups finalize within 2 weeks (**max 5** people). Proposal due early September, framed with the same five-step process.

3

Today's question

Once you actually have data sitting in front of you, how do you talk about it precisely enough to do anything useful with it?



You have seen data like this

Can you rank it? average it? or is it just a label?

T-shirt size	Phone number	Exam score	Movie genre
S, M, L, XL	*98765-43210*	*out of 100*	*comedy, horror, drama*

CORE CONCEPT — THE VOCABULARY STARTS HERE

Every dataset is objects and attributes

DATA OBJECT

An entity the dataset describes.

Also called: a sample, example, instance, or data point.

ATTRIBUTE

A measured property of that entity.

Also called: a feature, variable, or dimension.

Snow's data had exactly this shape: each cholera case was an object, recorded with symptoms, location, and water source as attributes.

A TINY DATASET

Student	Grade	GPA	Height (cm)
A. Rao	Sophomore	8.6	168
B. Iyer	Senior	7.9	175
C. Shah	Junior	9.1	162

Rows = objects. Columns (besides the name) = attributes.

GPA, and Height, they are numbers but feels different in kind.

The data matrix

N OBJECTS × D ATTRIBUTES

Every dataset is really just a matrix. Stack your N objects as rows, your D attributes as columns, and you get an N × D data matrix.

FEATURE VECTOR

Each row, once every attribute is encoded as a number, is what a model actually sees — one object's entire identity as a list of D numbers.

A CSV file is a data matrix — one line per object, one comma (delimiter) separated value per attribute.

OUR STUDENT TABLE, AS A MATRIX

Student	Grade	GPA	Height
A. Rao	**Sophomore**	**8.6**	**168**
B. Iyer	**Senior**	**7.9**	**175**
C. Shah	**Junior**	**9.1**	**162**

How many samples?

How many features?

Attribute types

CATEGORICAL

Nominal

A category, no meaningful order

Examples: blood type, payment mode

CATEGORICAL

Ordinal

A category, with meaningful order

Examples: Ratings

COUNTABLE

Discrete

Countable values

-1, 0, 10, 12, …

NUMERIC

Continuous

Any value, valid arithmetic

-0.45, 1.34, 21/2, π, …

CATEGORICAL

Nominal attributes

The attribute values has words, colours, pictures, or symbols used to represent a category, state, or label — with no meaningful order.

Examples

Blood type · Zip code · Payment mode · Phone number · Student ID

What CAN'T you do

No arithmetic operations are meaningful — you can't average a blood type, and shuffling the category labels loses nothing.

WHAT YOU CAN DO

Mode

The most commonly occurring value. It's the only measure of central tendency that survives on nominal data.

Nominal attribute(s) in the warm up?

T-shirt size	Phone number	Exam score	Movie genre
S, M, L, XL	*98765-43210*	*out of 100*	*comedy, horror, drama*

CATEGORICAL

Ordinal attributes

The attribute values has words, characters, or numbers that represents rank — they have meaningful order.

Examples

Movie rating (★,★★,…,★★★★★) · Education level · T-shirt size · Grade level

The test

Can you rank them, but not say “by how much”? If yes, it's ordinal.

WHAT YOU CAN DO

Median & Mode

For an odd number of values, the median (middle value) is meaningful because order is meaningful. Mode still works too.

Ordinal attribute(s) in the warm up?

T-shirt size	Phone number	Exam score	Movie genre
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DEBATE

Should you average a star rating (1 to 5)?

YES — AVERAGE IT

A restaurant with 4.5 stars from 200 reviews feels meaningfully better than one with 3.2. Averaging gives one comparable number, and it's what every review site actually does.

It's also fast to compute and trivial to sort by — one reason it won over more careful alternatives.

NO — DON'T AVERAGE IT

We don't know if the gap from 3★ to 4★ equals 4★ to 5★. A 4.5 average could hide a split crowd — half gave 5★, half gave 4★... or half gave 5★ and half gave 1★. The average erases that story.

A better fix: show the full distribution of 1★–5★ counts, not just one collapsed number.

Amazon, Flipkart shows both average and distribution. Ola, Uber shows average star ratings.

Discrete vs. continuous

Discrete

Countable values — words, characters, or numbers. Arithmetic isn't always meaningful.

EXAMPLES

-1, 0, 10, 12, …

WHAT YOU CAN DO

Mode. Sometimes median and mean, depending on context.

Continuous

Any value on a continuum. Valid arithmetic operations, always.

EXAMPLES

-0.45, 1.34, 21/2, π, …

WHAT YOU CAN DO

Everything — mode, median, mean, standard deviation, all of it.

PUTTING IT ALL TOGETHER

A real dataset, fully classified

A fitness app's weekly export. Five columns, four types, one table.

Attribute	Sample values	Type	Why
Workout type	Cardio, Strength, Rest	**Nominal**	*Categories with no natural order between them.*
Mood rating	1 – 5	**Ordinal**	*Ranked, but the gap from 2→3 isn't provably equal to 4→5.*
Resting heart rate	58, 61, 65 bpm	**Discrete**	*Whole numbers — countable beats per minute.*
Body weight	68.2, 68.5 kg	**Continuous**	*Any decimal value; full arithmetic is meaningful.*
Steps walked	8,420 / 10,502	**Discrete**	*Whole numbers — you can't walk half a step.*

Encodings: turning types into numbers

Ordinal

Assign integers

Freshman=1, Sophomore=2, Junior=3, Senior=4. Preserves order.

Discrete

One-hot encoding

Each value becomes its own 0/1 column — e.g. day-of-week as 7 flags instead of one number.

Nominal

Embedding

Each category maps to a learned vector — no false order is implied, unlike plain integers.



A CAUTIONARY TALE

When encodings go wrong

WRONG

One-hot encoding grade level

WRONG

Zip code stored as a plain integer

RIGHT

Match the encoding to the type

Grade level is ordinal → assign integers that preserve order (Freshman=1 … Senior=4). A model can then learn “higher is more senior” directly.

Zip code is nominal → one-hot or an embedding, never a raw integer. No false numeric relationship gets invented.

CLASSIFY IT! — POLL

Put your new vocabulary to work

For each attribute, vote: Nominal / Ordinal / Discrete / Continuous

1. **Jersey number**
2. **Satisfaction rating (1–5 stars)**
3. **Items in your cart**
4. **Body temperature (°C)**
5. **T-shirt size (S/M/L)**





WHY WE BOTHERED WITH THIS

Attribute type decides what math is valid

	Mode	Median	Mean	Std. Dev.
Nominal	✓	—	—	—
Ordinal	✓	✓	debated	—
Discrete	✓	✓	✓	✓
Continuous	✓	✓	✓	✓

Averaging a nominal attribute (like blood type) isn't just bad style — it's mathematically meaningless.

HALFWAY CHECKPOINT

So far, in one breath

1

Objects & attributes

Rows are Objects/Samples, Columns are Attributes/Features.

2

Four types, two families

Nominal & ordinal are categorical. Discrete & continuous are numeric.

3

Type decides valid math

Mode always works. Mean needs at least discrete data.

4

Encodings bridge to ML

Integers, embeddings, or one-hot — every type becomes numbers eventually.

BEFORE THE FORMULAS — A REALITY CHECK

Statistics is already part of your day

Highest / lowest marks, speeding, inflation rate

Maximum / Minimum

Average age, Expected arrival time

Mean

Stock market fluctuation

Variance

If it rains, there's more traffic

Correlation

Chance of rain tomorrow

Likelihood

Do people spend more on lunch or dinner, on average?

t-test

Mean — average

Let $x_1, x_2, \ldots, x_N$ be N values. The mean is:

$$X = \left(\frac{x_1 + x_2 + \cdots + x_N}{N}\right)$$

Weighted mean

If each value x_i carries a weight w_i, the weighted mean is

$$X = \left(\frac{\sum_{i=1}^{N} w_i x_i}{\sum_{i=1}^{N} w_i}\right)$$

TRY IT YOURSELF — 2 MINUTES

Compute a mean, live

THE DATA

A team's code-review turnaround times this week (in hours):

3, 5, 4, 22, 6

What is the mean?

Hold onto is 8 hours is a fair summary?

Once you have a number, ask yourself: does that number feel like a fair one-word summary?

When the mean lies

THE PROBLEM

Your code-review data had a 22 hiding in it. Now scale that idea up: 100, 200, 300, 1,000,000

Mean = 1,000,600 / 4 = 250,150

A single value dragged the "typical" number to 250,150 — nowhere near where three of the four values actually sit.

The mean is sensitive to outliers. Always.

THE FIX: TRUNCATED MEAN

Drop the extremes before averaging.

Values: -10000, 100, 200, 300, 1,000,000

Drop the min and max → (100+200+300)/3 = 200

Running time: sort first, so $O(Nlog\ N)$!!

Median, medoid, and mode

MEDIAN

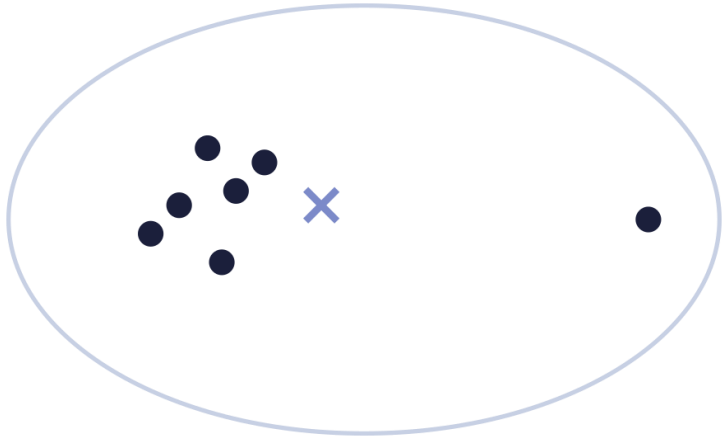
The middle value once sorted.
In the code-review data (3,4,5,6,22), the median is 5.

MODE

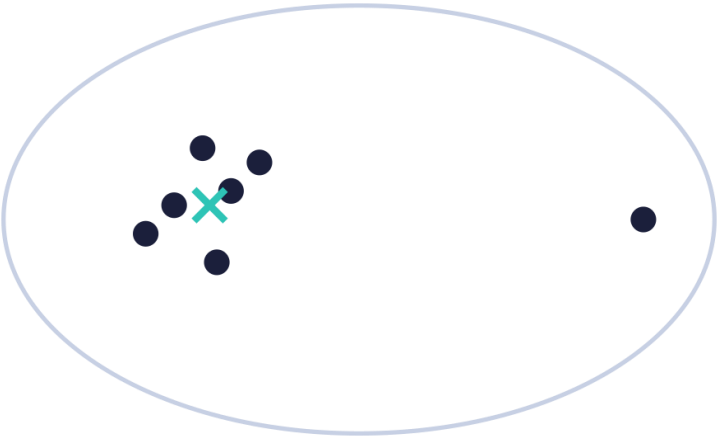
The most frequent value. Works on every attribute type — even nominal, where it's the only option.

MEDOID — median for higher dimensions

The point whose total distance to every other point is smallest — an actual data point, not an average.



Mean — pulled toward the outlier



Medoid — anchored in the real cluster

You can compute median in $O(N)$!!



TRY IT YOURSELF — 2 MINUTES

Compute your own weighted mean

Suppose your friend scored:

Assignment	**10%**	score: 90
Quiz	**20%**	score: 75
Mid-Sem	**20%**	score: 80
End-Sem	**25%**	score: 85
Project	**25%**	score: 70

Compute the final weighted grade.

DISCUSSION — 5 MINUTES

The dilemma

You run a typical IT company and want your report to show that a typical worker is paid well.

SCENARIO 1

A handful of employees earn very low salaries. As CEO, do you report the mean or the median salary? Why?

SCENARIO 2

Now suppose instead a lot more employees earn very low salaries, not just a handful. Does your answer change? Why?

Choosing a statistic is never neutral. The same dataset can support two honest-sounding but very different stories.

There's no single 'correct' answer here — the goal is to notice yourself making the choice.

QUICK QUIZ — COLD-CALL READY

Check your understanding

Q1 A dataset of 1000 salaries has one billionaire in it. The best measure of the 'typical' salary is:

Q2 Which attribute type can you meaningfully average?

Q3 One-hot encoding is the natural fit for which attribute type?

Reveal each answer only after a student attempts it out loud.

WRAP-UP

Before you go

Self Evaluate

1. Mean or median — which would you have picked as CEO, and why

2. How can we compute medoid

NEXT LECTURE

Describing & Cleaning Data

Dispersion, skewness, correlation, and the plots that reveal them — then how to actually clean the outliers, missing values, and noise you'll find in your project's data.

TAKEAWAY TO CARRY FORWARD

“The mean tells you a number. It doesn't tell you whether to trust it.”



STUDY MATERIAL

Reference

Data Mining: Chapter 2.1, 2.2 by Han, Kamber & Pei

Slides